

Lifelong Opportunity Guide 98 — Agricultural Technician

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Language: English

Primary U.S. occupation anchor: O*NET-SOC 19-4012.00 — Agricultural Technicians

Canada comparison: NOC 22110 — Biological technologists and technicians

Colombia comparison: CUOC 31421 — Técnicos agropecuarios

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What this career is

Agricultural technicians help scientists, agronomists, production teams, laboratories, growers, animal-production teams, seed operations, research organizations, government agencies, food/agricultural companies, and other professionals collect reliable information and carry out technical work in agriculture.

The work can combine field activities, laboratory work, crop or animal sampling, equipment setup, measurements, recordkeeping, controlled experiments, production monitoring, data entry and analysis, quality checks, irrigation or environmental measurements, seed work, research support, and equipment maintenance.

Job titles vary. Employers may use titles such as agricultural technician, agricultural research technician, agronomy research technician, field assistant, crop technician, seed technician, laboratory technician, research assistant, agricultural technologist, production technician, quality technician, precision-agriculture technician, or farm research technician.

This guide uses **O*NET-SOC 19-4012.00 — Agricultural Technicians** as its primary U.S. occupation anchor. The closest broad Canadian comparison used here is **NOC 22110 — Biological technologists and technicians**. The Colombian comparison is **CUOC 31421 — Técnicos agropecuarios**. These systems overlap, but they are not exact one-to-one equivalents.

An agricultural technician is not automatically an agronomist, agricultural scientist, veterinarian, licensed pesticide applicator, engineer, laboratory director, farm manager, or other regulated professional. Authority depends on the jurisdiction, employer, credential, task, product label, research protocol, and delegated responsibility.

Why this role can be an opportunity

Agricultural technology can suit people who like practical science, plants, animals, soils, field work, laboratories, measurements, equipment, data, and solving real production or research problems.

Transferable experience may come from:

- farming or ranching;
- gardening, horticulture, nurseries, or greenhouses;
- landscaping;
- food production;
- laboratory work;
- animal care;
- irrigation;
- environmental field work;
- GIS or mapping;
- machinery operation or maintenance;
- quality inspection;
- warehouse or inventory work;
- data collection;
- manufacturing;
- military technical work;
- community agriculture programs;
- agricultural sales or customer support.

A learner does not need to master every crop, animal species, laboratory technique, piece of equipment, or agricultural regulation. A strong entry foundation is narrower and more transferable: observe carefully, measure correctly, follow approved procedures, label samples accurately, protect data integrity, work safely, document deviations, respect regulated boundaries, and know when to stop and escalate.

What agricultural technicians may do

Depending on the employer and specialty, an agricultural technician may:

- assist scientists or agronomists with crop, soil, seed, fiber, pest, nutrition, breeding, or production research;
- collect field, crop, soil, water, feed, seed, tissue, or other approved samples;
- prepare specimens according to a written protocol;
- label samples and maintain traceability;
- set up approved laboratory or field equipment;
- check instrument status before use;

- record temperature, moisture, pH, mass, dimensions, counts, growth, yield, weather, irrigation, or other assigned variables;
- enter data into approved systems;
- review field sheets for missing or inconsistent entries;
- help establish experimental plots under a written design;
- monitor crop growth or production parameters;
- support seed-production or seed-quality activities;
- assist with greenhouse, nursery, irrigation, hydroponic, or controlled-environment work;
- support animal-production research or monitoring under approved animal-care procedures;
- prepare materials and equipment for tests;
- conduct assigned tests using approved methods;
- clean and maintain non-contaminated equipment within assigned authority;
- document equipment problems and request repair;
- assist with calibration checks where trained and authorized;
- support quality-control and good-production-practice records;
- photograph or map field observations where permitted;
- prepare charts, summaries, maps, or reports for professional review;
- support inventory control for samples, supplies, seed, tools, or approved chemicals;
- help investigate deviations, unusual results, or production problems;
- support precision-agriculture or sensor systems;
- communicate observations to scientists, supervisors, operators, growers, laboratory staff, or quality personnel.

Agricultural technician, scientist, agronomist, veterinarian, and farm worker are not the same role

Titles can overlap, but authority boundaries matter.

Agricultural technician

A technician generally performs defined technical work such as sampling, monitoring, measurement, testing, equipment support, field observations, controlled experimental tasks, data handling, and documentation.

Agricultural scientist or agronomist

A scientist or agronomist may design research, interpret complex biological or production systems, develop recommendations, make higher-consequence technical decisions, or hold credentials required by an employer or jurisdiction.

Veterinarian or veterinary professional

Animal-related agricultural work does not give a technician authority to diagnose disease, prescribe medication, select drug doses, determine withdrawal periods, or perform veterinary procedures outside delegated and legally permitted scope.

Pesticide applicator or pesticide handler

An agricultural-technician title does not automatically authorize mixing, loading, applying, disposing of, or working on contaminated pesticide equipment. Those activities can trigger separate training, label, employer, EPA Worker Protection Standard, state, local, or certification requirements.

Farm worker or equipment operator

Some agricultural technicians also perform production or equipment tasks. Others work mostly in research or laboratories. Never infer machine-operation authority from the technician title alone.

Core science foundations

Useful foundations include:

- basic biology;
- plant structure and growth;
- soil concepts;
- water and irrigation concepts;
- animal biology where relevant;
- ecology;
- chemistry fundamentals;
- measurement and units;
- percentages and ratios;
- basic statistics;
- experimental controls;
- contamination prevention;
- data integrity;
- safe equipment use.

You do not need to memorize every scientific term before applying. You do need enough foundation to recognize what you do not know and to follow a controlled method without inventing missing instructions.

Measurement and units

Agricultural data can become unusable when units are missing, converted incorrectly, or copied into the wrong field.

Build habits around:

- mass and weight;
- length and area;
- volume;
- temperature;
- time;
- moisture;
- pressure;
- flow;
- concentration as defined by an approved method;
- counts and rates;
- percentages;
- geographic coordinates;
- dates and timestamps.

Always record the unit required by the procedure or data system. Do not silently convert a result. If conversion is required, preserve the original value and document the conversion method when the procedure requires it.

Sampling and traceability

A technically perfect laboratory test cannot repair a mislabeled or contaminated sample.

A safe generic sampling workflow is:

1. read the approved sampling plan or protocol;
2. confirm the correct site, plot, animal group, material, date, and sampling window;
3. verify required containers, tools, labels, PPE, and documentation;
4. prevent cross-contamination using the approved cleaning or replacement method;
5. collect only the sample type and quantity specified by the method;
6. label immediately using the required identifier;
7. record time, location, conditions, collector, and other required metadata;
8. preserve/store/transport exactly as the approved method requires;
9. maintain chain-of-custody when required;
10. document deviations instead of hiding them;
11. deliver the sample to the designated person or location;
12. preserve the original record.

This guide intentionally does not specify pesticide, veterinary, microbiological, chemical, or regulated sampling methods. Use the employer's validated procedure or applicable regulator/laboratory method.

Field observations and research data

Strong field records separate what was **observed** from what is **inferred**.

A useful record can include:

- date and time;
- plot/site/sample identifier;
- observer;
- instrument ID;
- calibration or verification status when required;
- weather or environmental conditions when relevant;
- measurement and units;
- image/file identifier;
- unexpected condition;
- deviation from protocol;
- action taken;
- person notified.

Do not “clean up” a dataset by deleting unusual observations without an approved reason. An outlier may be an error, but it may also be the most important result in the experiment.

Laboratory basics

Agricultural technicians may work in seed, soil, plant, feed, environmental, biological, chemistry, quality, or production laboratories.

General laboratory habits include:

- read the current procedure before starting;
- identify required training and hazards;
- use the correct PPE;
- verify instrument readiness;
- check calibration/verification status when applicable;
- use clean and correctly identified containers;
- prevent cross-contamination;
- label reagents/materials according to the laboratory system;
- preserve raw results;
- record deviations;
- maintain housekeeping;
- dispose of waste through the approved route;

- report spills, exposures, breakage, equipment faults, or unexpected results.

Never invent a missing reagent concentration, incubation condition, sample-preservation rule, test limit, or acceptance criterion.

Crop and plant-production support

Depending on the job, a technician may support:

- crop establishment;
- plant growth monitoring;
- greenhouse operations;
- irrigation observations;
- propagation;
- seed production;
- harvest measurements;
- yield trials;
- variety trials;
- pest/disease scouting;
- soil sampling;
- nutrient-research trials;
- weed or insect research;
- nursery production;
- controlled-environment agriculture;
- field mapping.

Observation is not the same as prescription. A technician may record symptoms or pest counts without independently diagnosing a disease or prescribing a pesticide/fertilizer treatment unless that authority has been formally assigned and is legally permitted.

Animal-related agricultural work

Some agricultural technicians assist with animal breeding, nutrition, production, research, sample collection, or records.

Safe boundaries include:

- follow approved animal-handling and research protocols;
- use required restraint and PPE procedures;
- recognize species-specific behavioral hazards;
- report bites, scratches, sharps incidents, unusual illness, or suspected zoonotic exposure;
- maintain animal/sample identification;

- document feed, weight, production, environmental, or other assigned observations accurately;
- escalate suspected disease or welfare concerns to the responsible professional.

Do not diagnose illness, prescribe treatment, alter drug doses, change withdrawal periods, modify controlled diets, or perform procedures outside your authorized training and scope.

Pesticide work: agricultural worker and pesticide handler are different categories

The U.S. Environmental Protection Agency's Worker Protection Standard distinguishes agricultural **workers** from pesticide **handlers**.

Handler work can include activities such as:

- mixing or loading pesticides;
- transferring pesticides;
- applying pesticides;
- handling opened pesticide containers;
- cleaning, adjusting, or repairing contaminated application equipment;
- assisting with pesticide application;
- certain crop-advisor entry tasks;
- disposing of pesticides or pesticide containers.

An agricultural-technician job title does not by itself make you an authorized pesticide handler or applicator.

Pesticide safety: follow the label and the regulated program

The pesticide label, applicable law, employer program, Worker Protection Standard, training, certification, PPE, application exclusion zones, restricted-entry intervals, decontamination requirements, and emergency procedures can be legally consequential.

This guide therefore does **not** provide:

- pesticide rates;
- tank-mix recipes;
- dilution instructions;
- off-label uses;
- restricted-entry interval calculations;
- re-entry authorization;
- respirator selection;
- application method instructions;

- disposal instructions for a specific pesticide;
- recommendations to treat a crop, animal, pest, or disease.

If a pesticide task is assigned, use the current product label and the employer's approved process and confirm that you have the required training, certification, PPE, medical evaluation/fit testing when required, and authority.

Agricultural machinery and PTO safety

Agricultural work can involve tractors, harvesters, augers, conveyors, pumps, irrigation equipment, mowers, tillage equipment, grain systems, shop equipment, and power take-off (PTO) driven machinery.

OSHA agriculture rules require guarding of specified power-transmission hazards. Practical technician rules include:

- keep required guards and shields in place;
- do not reach across or step over a rotating shaft;
- keep loose clothing, hair, jewelry, cords, and tools away from rotating machinery;
- make sure people are clear before equipment starts or power is engaged;
- do not service a machine merely because the normal control is off;
- follow the approved energy-control/isolation process before service or maintenance;
- do not work under unsupported raised equipment;
- inspect/report missing or damaged guards;
- use only equipment for which you are trained and authorized;
- follow manufacturer and employer instructions.

A PTO can cause catastrophic entanglement. Never treat it as a routine moving shaft that can be approached because it appears slow.

Tractor and vehicle boundaries

Where tractor work is part of the job:

- use required roll-over protection and seat belts where applicable;
- follow operator training requirements;
- inspect equipment as required;
- do not carry unauthorized passengers;
- understand slopes, loads, attachments, visibility, braking, and stability limits;
- stop and report unsafe equipment;

- follow employer rules for roads, trailers, ATVs/UTVs, and mobile equipment.

This guide is not an operator-certification course.

Heat, weather, and field conditions

Field work may expose technicians to heat, sun, storms, cold, uneven ground, insects, animals, dust, allergens, noise, and remote work locations.

Follow the employer's site-specific controls. In hot conditions, common program elements include hydration, rest/cooling or shade, acclimatization, symptom recognition, supervision, and emergency response.

Pesticide-required PPE may increase heat burden. Do not remove required PPE because it is uncomfortable; instead follow the employer's heat and pesticide-safety process and escalate unsafe conditions.

Biological, zoonotic, and sharps hazards

Agricultural research can involve animals, tissues, biological samples, needles, insects, microorganisms, manure, soil, wastewater, or other biological material.

Use the biosafety and animal-care controls assigned by the employer or institution. Report bites, scratches, needlesticks, splashes, spills, suspected zoonotic exposure, or unusual illness promptly.

Do not assume a sample is harmless because it comes from a farm rather than a hospital laboratory.

Chemical safety beyond pesticides

Agricultural work may involve fertilizers, fuels, cleaning products, laboratory chemicals, disinfectants, acids/bases, compressed gases, or other hazardous products.

General rules:

- identify the product before use;
- read the current Safety Data Sheet and label when required;
- use the approved container;
- never use an unlabeled chemical;
- follow storage compatibility rules;
- use required ventilation and PPE;
- do not combine products unless an approved procedure explicitly permits it;
- know spill/emergency reporting routes;
- keep food and drink out of prohibited work areas.

Irrigation, water, and environmental monitoring

Technicians may support irrigation or drainage systems and collect data such as flow, pressure, moisture, weather, water quality, or field conditions.

Use calibrated/verified equipment when the procedure requires it. Do not change pump, pressure, electrical, chemical-injection, or water-treatment settings outside assigned authority.

Environmental or discharge compliance decisions should be escalated to the responsible professional when outside the technician's delegated scope.

Precision agriculture, GIS, sensors, and drones

Modern agricultural operations may use:

- GPS/GNSS;
- GIS;
- remote sensing;
- yield monitors;
- weather stations;
- soil-moisture sensors;
- connected irrigation;
- machine telemetry;
- cameras;
- drones;
- cloud dashboards;
- field-data platforms.

Learn the transferable skills: coordinate systems, metadata, sensor identification, calibration/verification status, data synchronization, file naming, quality checks, backup, and change control.

Drone operation can require separate aviation authority, training, registration, operational, airspace, or employer requirements. Do not assume that an agricultural-technician job title authorizes drone flight.

Data integrity and version control

Agricultural decisions can be distorted by weak records.

Protect:

- sample IDs;
- plot IDs;
- animal IDs;
- dates/times;
- units;

- raw measurements;
- treatment codes;
- instrument IDs;
- calibration status;
- protocol versions;
- map layers;
- images;
- laboratory results;
- chain-of-custody records;
- approved corrections.

When correcting an entry, follow the approved audit-trail method. Do not overwrite or backdate a record to make a trial, inspection, or production result look better.

Cybersecurity in connected agriculture

Agricultural and research systems increasingly connect physical operations to digital systems.

Good practices include:

- use employer-approved devices and accounts;
- use MFA where available;
- protect credentials and recovery codes;
- do not connect unknown USB devices to lab, field, or production systems;
- verify unusual requests to install software or firmware;
- follow controlled update and backup procedures;
- separate personal accounts/devices from work systems where required;
- protect proprietary crop, farm, customer, location, research, animal, and production data;
- report phishing, malware, account compromise, or unexplained connected-equipment behavior;
- do not share remote-access credentials with a vendor outside the approved process.

Cybersecurity can become a safety and production issue when connected equipment affects irrigation, environmental controls, machinery, or laboratory systems.

Responsible AI in agricultural technical work

AI can help with low-risk learning and drafting when the employer or school allows it.

Reasonable uses may include:

- explaining a public scientific concept in simpler language;
- generating fictional practice questions;
- suggesting headings for a non-confidential report;
- improving grammar;
- organizing public research;
- creating a generic checklist draft for human review;
- helping write spreadsheet formulas using fictional data.

Human verification is required for consequential technical work.

Do not rely on AI as final authority for:

- pesticide rates, tank mixes, label directions, restricted-entry intervals, or application decisions;
- fertilizer or chemical prescriptions;
- veterinary diagnosis, medication, dose, or withdrawal-period decisions;
- crop disease or pest diagnosis without qualified verification;
- animal welfare or treatment decisions;
- laboratory methods or acceptance criteria;
- sample results or field observations;
- machinery guarding or safe-operation decisions;
- calibration acceptance;
- irrigation chemical-injection decisions;
- regulated environmental decisions;
- research conclusions;
- food/feed safety release decisions;
- legal or regulatory compliance.

Never use AI to fabricate a sample result, field observation, treatment record, animal record, chain-of-custody entry, laboratory result, calibration record, or experimental outcome.

Do not upload confidential research, proprietary farm/customer information, credentials, regulated records, unpublished findings, precise sensitive location information, or other protected data to an unapproved AI service.

Education and entry paths — United States

BLS describes postsecondary education, commonly an **associate degree**, as a typical route for agricultural and food science technician work, although requirements vary by employer and specialty.

Relevant programs can include:

- agricultural science;
- agricultural technology;
- agronomy;
- plant science;
- animal science;
- soil science;
- horticulture;
- precision agriculture;
- biotechnology;
- laboratory technology;
- environmental science/technology;
- GIS;
- irrigation technology;
- farm equipment technology.

Before paying for a program, compare:

- total tuition and fees;
- required field/lab equipment;
- laboratory access;
- supervised field experience;
- internship/co-op options;
- transfer pathways;
- employer partnerships;
- pesticide-related training boundaries;
- animal-care training where relevant;
- current placement evidence;
- schedule and transportation requirements.

U.S. training and funding locator

CareerOneStop provides a WIOA-eligible training-program locator:

<https://www.careeronestop.org/LocalHelp/EmploymentAndTraining/find-WIOA-training-programs.aspx>

An American Job Center can explain current local options. Eligibility, approved providers, supportive services, tuition assistance and other funding vary by workforce area and individual circumstances.

Search related program families, not only “agricultural technician.”

No funding approval is guaranteed by this guide.

Canada pathway

Job Bank maps agricultural technician to **NOC 22110 — Biological technologists and technicians**.

The NOC family supports professionals in agriculture, natural-resource management, environmental protection, plant and animal biology, microbiology, biotechnology and related laboratory/field settings.

Job requirements vary. College training is a common route, with technician and technologist pathways differing in length and depth. Always check the current province, employer, title, and credential expectations.

Canada wages and outlook

Job Bank national wages, updated **November 19, 2025**, are:

- low: **C\$19.50/hour**;
- median: **C\$29.12/hour**;
- high: **C\$43.00/hour**.

The reference period is 2023-2024 and the source is the Labour Force Survey / Statistics Canada.

Job prospects vary by province and territory. Do not treat the number of currently advertised Job Bank postings as a forecast.

Canada training and employment supports

Start with:

<https://www.canada.ca/en/services/jobs/training.html>

Government of Canada Labour Market Agreements provide a major route through which provinces and territories deliver training, upskilling and employment supports:

<https://www.canada.ca/en/employment-social-development/programs/training-agreements.html>

Labour Market Development Agreement information is available at:

<https://www.canada.ca/en/employment-social-development/programs/training-agreements/lmda.html>

Eligibility and available support differ by jurisdiction and individual circumstances. These are locators, not promises of payment or admission.

Colombia pathway — CUOC 31421

OCUPACOL identifies **CUOC 31421 — Técnicos agropecuarios**, competency level 3.

The profile includes work supporting agricultural, biological, production, seed, laboratory and resource-management professionals. Functions include specimen collection, sampling/testing, preparation of materials/equipment, experimental support, measurement/monitoring, analysis/reporting, good-production-practice activities and production support.

The CUOC classification is a strong occupational reference, but the exact job title and employer scope still matter.

Colombia wage-data caution

OCUPACOL currently displays approximately **COP 1,509,958-4,560,530** for this occupation, but the page explicitly states that the data **do not have statistical representativeness under the applied methodology**.

Therefore:

- do not present this as a statistically representative current national salary;
- keep the caveat beside the number if the range is shown;
- check current employer vacancies and contract conditions;
- separate job-ad examples from official labor-market statistics.

SENA route — Gestión de la Producción Agrícola

SENA Betowa lists **Gestión de la Producción Agrícola** as:

- Titulada;
- **Tecnólogo**;
- **3,984 hours**;
- **Presencial**;
- age **15+** on the reviewed page;
- state secondary-education exam result required on the reviewed offer;
- admission subject to available seats and SENA selection.

Competencies include production control, agricultural data, irrigation/drainage, crop establishment, harvesting, propagation, formative research, production planning, environmental/occupational-safety practices, quantitative reasoning and digital tools.

Cohort, location, date, employer-sponsored status and seat availability can change. Recheck Betowa before planning enrollment.

SENA route — Cultivos Agrícolas

SENA Betowa lists **Cultivos Agrícolas** as:

- Titulada;
- **Técnico**;
- **2,200 hours**;
- **Presencial**;
- age **16+**;
- 9th grade completed on the reviewed page;
- additional selection/identity requirements;
- admission subject to seats and SENA procedure.

The program covers soil preparation, planting, crop maintenance, harvesting, good agricultural practices and cleaner technologies.

Keep this 2,200-hour technician route distinct from the 3,984-hour technologist route.

Latin America and Caribbean training locator

OIT/Cinterfor can help locate national vocational-training institutions:

<https://www.oitcinterfor.org/>

<https://www.oitcinterfor.org/statsfp/paises>

Search national systems for agricultural technology, crop production, agronomy support, animal production, horticulture, irrigation, precision agriculture, laboratory technology, environmental monitoring, machinery, GIS and related fields.

Verify current admission, cost, modality, credential, funding and employer recognition directly with the institution.

U.S. official pay picture

O*NET's updated 2026 page uses 2025 BLS wage data and reports:

- **\$23.86/hour median**;
- **\$49,630/year median**;
- 10th percentile **\$35,980/year / \$17.30/hour**;
- 25th percentile **\$40,950/year / \$19.69/hour**;
- 75th percentile **\$63,630/year / \$30.59/hour**;
- 90th percentile **\$76,480/year / \$36.77/hour**.

It reports 18,600 workers in 2024 and average projected growth of 3%–4% over 2024–2034.

Earlier BLS OOH pay and projection context

The BLS Occupational Outlook Handbook reports an agricultural-technician median of **\$46,790/year in May 2024**, with the lowest 10% below **\$32,860** and highest 10% above **\$69,010**.

BLS projects agricultural-technician employment from **18,600 in 2024 to 19,400 in 2034**, approximately **+800 / 4%**.

Do not substitute the broader Agricultural and Food Science Technicians group's figures for the agricultural-technician-only numbers.

Non-government U.S. market context

ZipRecruiter reported, as of **August 7, 2026**, an average around:

- **\$43,987/year**;
- **\$21.15/hour** equivalent;
- approximately **\$37,000-\$41,500** for the 25th-75th percentile annual range;
- approximately **\$57,500** at the 90th percentile.

Source: <https://www.ziprecruiter.com/Salaries/Agricultural-Technician-Salary>

This is a private platform estimate, not a BLS statistic. Use it only as secondary market context.

Current Colombia job-ad context

Computrabajo currently lists agricultural-technician-related vacancies using titles such as Técnico Agropecuario, Técnico Agrícola/Agropecuario and Tecnólogo Agropecuario. Recent indexed examples span roughly **COP 1.75M-3.54M/month**, depending on the role and location.

Source/locator: <https://co.computrabajo.com/trabajo-de-tecnico-agropecuario>

These are transient advertisements, not a representative national wage survey. Some jobs include sales, animal-production, field-service or supervisory duties that are not equivalent to the core CUOC 31421 profile.

Accessibility and work fit

Agricultural technician jobs differ greatly in physical demands.

Some are mainly:

- laboratory work;
- sample preparation;

- data entry/analysis;
- GIS or mapping;
- seed quality;
- documentation;
- controlled-environment monitoring.

Others include:

- walking fields;
- uneven terrain;
- heat/sun;
- rain or cold;
- lifting within assigned limits;
- ladders or greenhouse work;
- animal exposure;
- machinery noise;
- dust/allergens;
- travel;
- early-morning or seasonal schedules.

Read the actual essential job functions. Use the employer or school accommodation process when needed. This guide does not determine legal entitlement to any particular accommodation.

For learning accessibility, consider:

- short study blocks;
- step-by-step checklists;
- diagrams paired with text explanations;
- consistent units;
- labeled examples;
- digital notes;
- text-to-speech or speech-to-text;
- supervised practical demonstrations;
- repeated practice with fictional or safe non-regulated examples.

Safe portfolio ideas

A portfolio should demonstrate technical thinking without pretending to provide regulated agricultural advice.

Possible projects include:

- a fictional crop-trial data table and chart;
- a sample-label and chain-of-custody exercise using fictional samples;
- a field-observation form;

- a GIS map using public or fictional data;
- a moisture/temperature sensor dataset using safe demonstration equipment;
- a calibration-status tracking sheet;
- a generic equipment-inspection checklist that tells the user to follow the manufacturer/employer procedure;
- an experimental-plot layout exercise clearly labeled educational;
- a data-quality audit showing missing units, duplicate IDs, or inconsistent dates;
- a mock research report using entirely fictional results.

Do **not** present a portfolio pesticide recommendation, veterinary treatment, fertilizer prescription, real-world disease diagnosis, or fabricated field/lab result as professional work.

Six-step action plan

This controlled edition repairs the legacy Action Plan 1 To 6: False defect. Each step has a measurable milestone.

Step 1 — Choose a target agricultural-technician track

Actions:

- review at least 10 current job descriptions;
- compare crop, seed, soil, laboratory, animal-production, research, precision-agriculture, irrigation, and production roles;
- record repeated education, safety, equipment, data, schedule, and physical requirements;
- choose one realistic first target.

Milestone: complete a one-page role comparison and select one primary target plus one backup target.

Step 2 — Build the science, measurement, and data foundation

Actions:

- review basic biology, chemistry, algebra, units, percentages, and basic statistics;
- practice sample IDs, units, timestamps, and data-entry quality checks;
- learn contamination-prevention and traceability concepts;
- review the difference between observation and professional recommendation.

Milestone: score at least 80% on a 20-question self-test and correct every missed item.

Step 3 — Complete a relevant training path

Actions:

- compare current college, technical, SENA, employer, or national vocational programs;
- verify cost, duration, modality, field/lab access, safety training, admission, and internship/co-op options;
- check WIOA, Canadian provincial/territorial supports, SENA, or other local funding locators when applicable.

Milestone: create a written training plan showing the selected program/path, prerequisites, start window, total known cost, and funding questions still to verify.

Step 4 — Build safe supervised technical practice

Actions:

- practice sample labeling and fictional chain-of-custody records;
- use safe measurement tools under supervision where appropriate;
- practice field/lab documentation;
- learn pesticide-handler boundaries, machinery guarding/PTO hazards, heat controls, chemical safety, animal/biosafety boundaries, and stop-work escalation;
- build one non-hazardous portfolio project.

Milestone: complete a safety-and-scope checklist with at least 12 hazards/boundaries and identify the correct escalation owner for each.

Step 5 — Obtain supervised experience and evidence

Actions:

- pursue an internship, co-op, trainee role, supervised research, farm/lab placement, seed operation, greenhouse, production project, or approved school laboratory;
- keep non-confidential notes on tasks, tools, measurements, procedures, problems, and results;
- request feedback on accuracy, safety, documentation, reliability, and teamwork.

Milestone: document three truthful achievement examples using situation, action, and result, with no confidential or fabricated data.

Step 6 — Apply, interview, and improve

Actions:

- tailor the resume to the selected technician track using only supportable evidence;
- prepare examples involving sampling, measurement, data integrity, safety, field/lab teamwork, and escalation;
- apply consistently to realistic roles;
- track applications, responses, interviews, and skill gaps;
- revise the plan using evidence rather than guesswork.

Milestone: submit 10 well-aligned applications, complete two mock interviews, and identify the next two evidence-based improvements from the tracker.

Interview preparation

Be ready to explain:

- how you prevent sample-labeling errors;
- what you do when a measurement seems wrong;
- why units and timestamps matter;
- how you document a protocol deviation;
- what you do if a pesticide task is assigned and you are not sure you are authorized;
- why a PTO or unguarded shaft is dangerous;
- what you do before servicing equipment;
- how you distinguish observation from diagnosis or treatment advice;
- how you protect research or farm data;
- how you use AI without trusting it for regulated or safety-critical decisions.

Strong answers are specific, careful, evidence-based, and honest about scope.

Final checklist

Before applying, confirm that you can honestly say:

- ☐ I can explain what an agricultural technician does.
- ☐ I understand the difference between technician support and scientist/agronomist/veterinary authority.
- ☐ I understand that technician title alone does not authorize pesticide-handler/applicator work.
- ☐ I can label and track a fictional sample correctly.
- ☐ I can record measurements with correct units and timestamps.
- ☐ I understand why raw data, deviations, and traceability matter.
- ☐ I understand PTO, guarding, and machinery-service hazards.
- ☐ I know when to stop work and escalate.

- ☐ I understand basic chemical and biological safety boundaries.
- ☐ I can explain safe, limited AI use in agricultural technical work.
- ☐ I can protect research, farm, customer, and credential data.
- ☐ I checked current education and funding options in my location.
- ☐ I have at least one safe, truthful portfolio or supervised-experience example.
- ☐ I completed all six action-plan milestones or assigned dates to finish them.

Sources and verification links

United States

- O*NET OnLine — Agricultural Technicians:
<https://www.onetonline.org/link/details/19-4012.00>
- U.S. Bureau of Labor Statistics — Agricultural and Food Science Technicians:
<https://www.bls.gov/ooh/life-physical-and-social-science/agricultural-and-food-science-technicians.htm>
- CareerOneStop WIOA-Eligible Training Program Finder:
<https://www.careeronestop.org/LocalHelp/EmploymentAndTraining/find-WIOA-training-programs.aspx>
- EPA Agricultural Worker Protection Standard:
<https://www.epa.gov/pesticide-worker-safety/agricultural-worker-protection-standard-wps>
- EPA Definition of a Pesticide Handler: <https://www.epa.gov/pesticide-worker-safety/definition-pesticide-handler-under-worker-protection-standard>
- EPA Restrictions During Pesticide Applications:
<https://www.epa.gov/pesticide-worker-safety/restrictions-during-pesticide-applications-under-worker-protection-standard>
- OSHA Agriculture standards:
<https://www.osha.gov/laws-regs/regulations/standardnumber/1928/1928TableofContents>
- OSHA 1928.57 — farm equipment guarding:
<https://www.osha.gov/laws-regs/regulations/standardnumber/1928/1928.57>
- OSHA Agricultural Operations — Hazards & Controls:
<https://www.osha.gov/agricultural-operations/hazards>
- NIOSH Agriculture Worker Safety and Health:
<https://www.cdc.gov/niosh/agriculture/about/>
- CISA Secure Our World: <https://www.cisa.gov/secure-our-world>
- NIST AI Risk Management Framework: <https://www.nist.gov/itl/ai-risk-management-framework>

- NIST Generative AI Profile:
<https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>

Canada

- Job Bank Agricultural Technician summary:
<https://www.jobbank.gc.ca/marketreport/summary-occupation/3088/ca>
- Job Bank Agricultural Technician wages:
<https://www.jobbank.gc.ca/marketreport/wages-occupation/3088/ca>
- Job Bank Agricultural Technician requirements:
<https://www.jobbank.gc.ca/marketreport/requirements/3088/ca>
- Government of Canada training:
<https://www.canada.ca/en/services/jobs/training.html>
- Labour Market Agreements: <https://www.canada.ca/en/employment-social-development/programs/training-agreements.html>
- Labour Market Development Agreements:
<https://www.canada.ca/en/employment-social-development/programs/training-agreements/lmda.html>

Colombia and Latin America

- OCUPACOL CUOC 31421:
<https://ocupacol.mintrabajo.gov.co/Profile/OccupationalProfile/31421>
- OCUPACOL hierarchy:
<https://ocupacol.mintrabajo.gov.co/Jerarquias/Jerarquias>
- SENA — Gestión de la Producción Agrícola:
<https://betowa.sena.edu.co/oferta/gestion-de-la-produccion-agricola?location=57054001&modality=P&offertype=company&programId=101813>
- SENA — Cultivos Agrícolas:
<https://betowa.sena.edu.co/oferta/cultivos-agricolas?modality=P&programId=66521>
- OIT/Cinterfor: <https://www.oitcinterfor.org/>
- OIT/Cinterfor country network:
<https://www.oitcinterfor.org/statsfp/paises>

Secondary market context

- ZipRecruiter Agricultural Technician salary:
<https://www.ziprecruiter.com/Salaries/Agricultural-Technician-Salary>
- Computrabajo Colombia agricultural-technician listings:
<https://co.computrabajo.com/trabajo-de-tecnico-agropecuario>

Important notice

This guide provides general educational and career-planning information. It is not agronomic, veterinary, pesticide, engineering, laboratory,

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